1. Problem

AICTE receives institution-related complaints through multiple Bureaus and Cells, including the Regulation Bureau (RB) and Public Grievance Redressal Cell (PGRC). Since cases may be handled independently, related complaints can remain disconnected, resulting in duplicated work, delayed action, unclear ownership, and limited visibility into case progress.

- Fragmented monitoring: No consolidated view of an institution's complaints across Bureaus/Cells.
- Overlapping complaints: Different departments may process complaints about the same or related incidents without coordination.
- Unclear accountability: Difficulty identifying the responsible Bureau, officer, next action, and expected disposal date.
- Delayed resolution: Pending cases, missed deadlines, and unresolved departmental dependencies may not receive timely attention.
- Limited analytics: Consolidating institution-wise, region-wise, and Bureau-wise reports can be difficult.

Objective: Develop a centralized institution complaint monitoring and coordination dashboard that improves visibility, identifies potential overlaps, tracks responsibility, and supports timely action across AICTE departments.

2. Solution

AICTE Unified Grievance Coordination and Accountability Dashboard

A centralized platform that integrates institution-related case information across AICTE Bureaus and Cells, combining all mandatory SIH monitoring features with an AI-assisted Cross-Bureau Coordination Engine.

Mandatory features

- Centralized institution profiles: AICTE Permanent ID, institution details, region, and consolidated complaint history.
- Complete case tracking: Unique case ID, receipt date, category, status, assigned Bureau/Cell, responsible officer, last and next action, and expected disposal date.
- Cross-Bureau overlap detection: Identify potentially related cases using institution identity, complaint text, category, and incident context.
- Real-time dashboard: Monitor received, pending, under-process, disposed, transferred, escalated, and overdue cases.

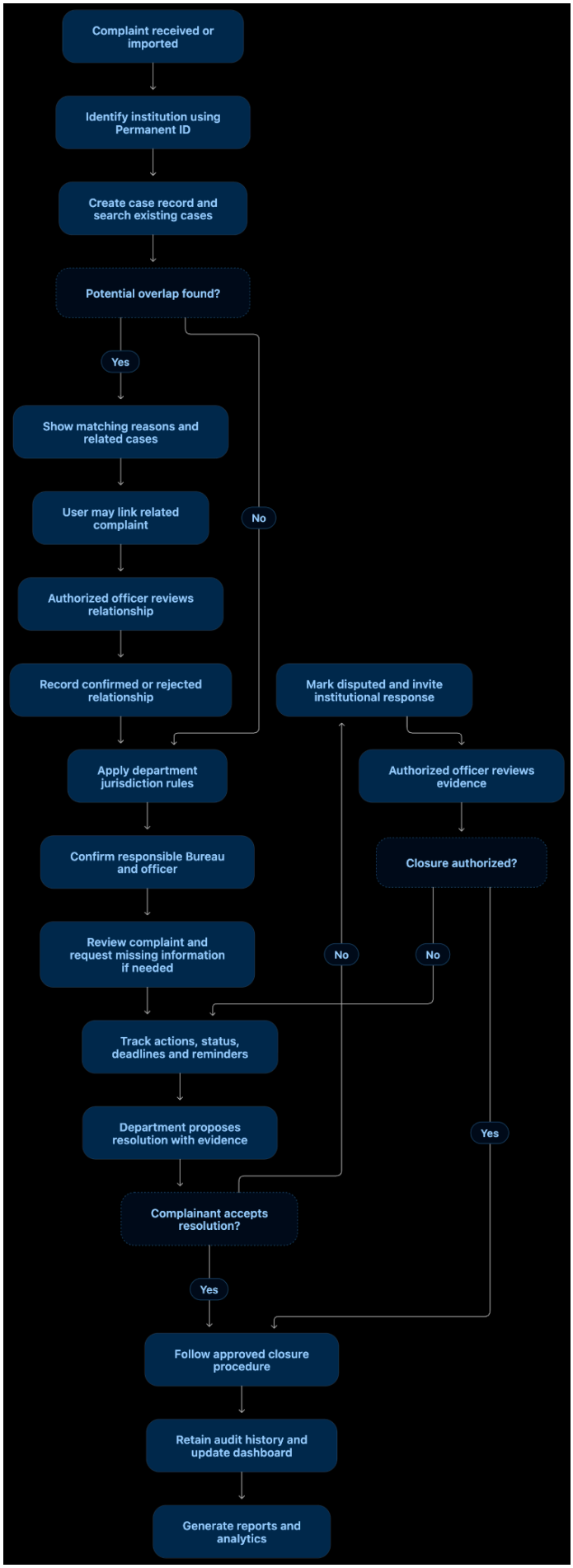
- Chronological action history: Record actions, timestamps, remarks, assignments, transfers, and supporting-document references.
- Search and filters: Find cases by institution, Permanent ID, case ID, Bureau/Cell, region, category, status, and date range.
- Alerts and notifications: Highlight potential overlaps, pending actions, approaching deadlines, and overdue cases.
- Reports and analytics: Generate Bureau-wise, region-wise, and institution-wise reports, ageing analysis, disposal trends, and overlap statistics.

Core USP — AI-Assisted Cross-Bureau Coordination and Accountability Engine

- Smart complaint linking: Suggest related unresolved complaints and allow users to link relevant cases, subject to the portal's permissions and workflow.
- Controlled assignment: Support department claims for unassigned cases, with jurisdiction checks and authorized confirmation.
- Two-sided resolution verification: Allow complainants to dispute proposed resolutions and institutions or departments to submit supporting evidence.
- Action accountability: Track responsible officers, next actions, deadlines, transfers, and escalations.
- Human-reviewed decisions: Preserve individual complaint histories and require authorized review for disputed outcomes and suspected misuse.

Key distinction: The system does not merely display cases or detect similarity. It provides a traceable process for reviewing overlaps, coordinating departments, and tracking outcomes while preserving each complaint's identity.

3. Use flow / Workflow


Complaint received or imported
Identify institution using Permanent ID
Create case record and search existing cases
Potential overlap found?
Yes
No
Show matching reasons and related cases
User may link related complaint
Authorized officer reviews relationship
Record confirmed or rejected relationship
Mark disputed and invite institutional response
Apply department jurisdiction rules
Authorized officer reviews evidence
Confirm responsible Bureau and officer
Closure authorized?
Review complaint and request missing information if needed
No
No
Track actions, status, deadlines and reminders
Department proposes resolution with evidence
Yes
Complainant accepts resolution?
Yes
Follow approved closure procedure
Retain audit history and update dashboard
Generate reports and analytics

Workflow in simple terms

1. Receive a complaint or import an existing case through an authorized process.
2. Identify the institution and search for potentially related complaints.
3. Display overlap alerts with reasons; allow appropriate linking and officer review.
4. Determine the responsible Bureau and officer using approved routing rules.
5. Track information requests, actions, status changes, deadlines, and escalation.
6. Record the proposed resolution and supporting evidence.
7. Allow the complainant to confirm or dispute the proposed resolution.
8. Give the institution an opportunity to respond to disputes; an authorized officer determines the outcome under applicable procedures.
9. Update the dashboard, retain the complete action history, and generate reports.

Important safeguards: Linked complaints retain separate IDs, histories, and applicable deadlines. An OTP verifies participation in an action, not whether the underlying issue is actually resolved. The two-day response period, penalties, and false-complaint thresholds remain optional, policy-dependent enhancements rather than assumed AICTE rules.

4. Feasibility / Integration

Technical feasibility

Component	Proposed implementation
Frontend	React, TypeScript, Tailwind CSS
Database and backend	Supabase and PostgreSQL
Authentication	Role-based access for authorized users
Complaint matching	PostgreSQL full-text search and `pg_trgm` similarity
AI enhancement	Optional semantic similarity model after initial validation

Dashboard and reports	Recharts and downloadable reports
Notifications	In-app alerts, reminders, and configurable escalation
Evidence management	Private file storage and permission-controlled access

Integration approach

- Institution registry: Import authorized institution data using the AICTE Permanent ID as the principal identifier.
- RB, PGRC, and other Bureaus/Cells: Map records into a common case structure while retaining the original department and case reference.
- Existing complaint systems: Begin with authorized data imports; use APIs or synchronization only when access is approved.
- Authentication and permissions: Restrict case details and evidence to authorized roles.
- Reports and alerts: Calculate case counts, ageing, deadlines, and workload from stored case and activity records.

Feasibility, risks, and mitigation

Challenge	Mitigation
False or missed overlap matches	Test against labelled examples; use explainable alerts and human review
Incorrect case assignment	Apply jurisdiction rules and authorized confirmation
Unavailable internal AICTE data	Demonstrate with clearly labelled synthetic data

Incomplete status updates	Record timestamps, responsible officers, and overdue actions
Disputed resolution	Support evidence submission, institutional response, and authorized review
Privacy and security	Enforce role-based access, secure evidence storage, and audit logging
Existing-system integration	Use modular connectors and approved imports before live integration

Viability

The solution is feasible as a prototype using established web technologies. Production deployment depends on access to reliable AICTE data, authorization to integrate existing systems, and approved procedures for assignment, deadlines, coordination, and closure.

Success metrics:

- Precision and recall of overlap detection.
- Time required to identify related complaints.
- Percentage of cases with a responsible officer and next action.
- Overdue-case rate and escalation compliance.
- Number of confirmed overlaps leading to documented coordination.
- Reopened or disputed resolution rates.

Final positioning: The mandatory SIH requirements form the foundation; the AI-assisted coordination engine is the core USP; and user-side linking, two-sided resolution verification, and additional accountability rules are enhancements built around that foundation.

IMP NOTE: Add this under Solution – Core USP and include it in the workflow. The + button is your user-facing innovation.

Smart Complaint Linking — "+ Link to Existing Complaint"

When a complainant files a new complaint, the system searches for similar unresolved complaints associated with the same institution. If potential matches are found, it displays them to the complainant with a "+ Link to Existing Complaint" button.

- View matches: Show existing complaint IDs, summaries, statuses, and responsible Bureaus where the user is authorized to view them.
- Link with one click: The complainant can select the + button to indicate that their complaint may be related to an existing case.
- Preserve separate records: The new complaint retains its unique ID, history, and ownership; linking does not automatically merge or close either complaint.
- Verify the relationship: The system flags the link for review and allows authorized officers to confirm or reject it.
- Track progress: Once linked, the complainant can track their own complaint and receive permitted updates, subject to access and privacy rules.

Example: A student files a complaint about an institute withholding original certificates. The system finds a similar unresolved complaint for the same institute and displays:

Similar complaint found

Example data for demonstration

Case PGRC-1042

Original certificates withheld · Under process

+ Link to Existing Complaint

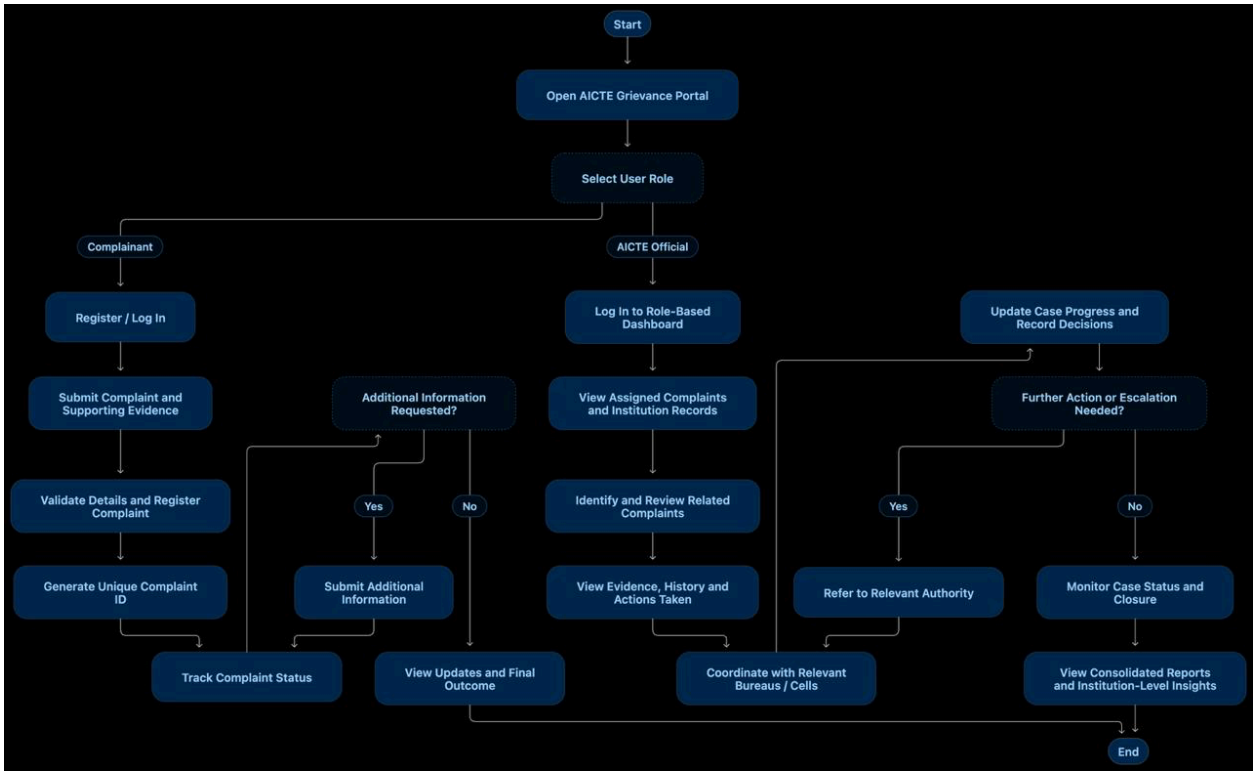
Indicates a possible relationship for review; does not merge cases.

Add this to your workflow

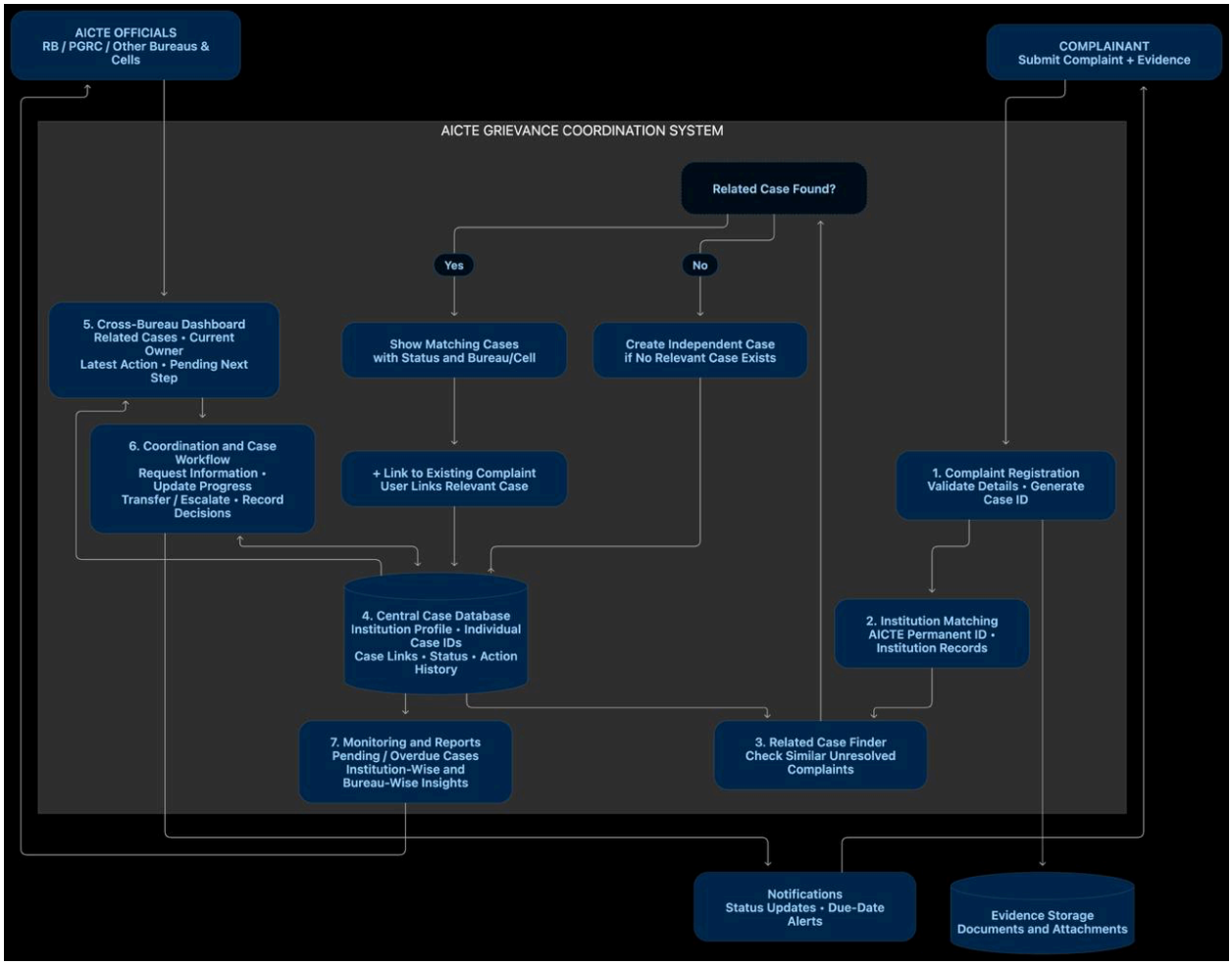
Complaint submitted → Similar unresolved complaints detected → Matching cases displayed → User clicks "+ Link to Existing Complaint" → Relationship reviewed by authorized officer → Cases linked if confirmed → Individual status tracking continues.

This feature complements the mandatory cross-Bureau overlap detection and case-linking requirements while adding a simple way for complainants to help identify potentially related cases.

User flow:



System architecture:


AICTE OFFICIALS
RB / PGRC / Other Bureaus & Cells
COMPLAINANT
Submit Complaint + Evidence
AICTE GRIEVANCE COORDINATION SYSTEM
Related Case Found?
Yes
No
5. Cross-Bureau Dashboard
Related Cases • Current Owner
Latest Action • Pending Next Step
Show Matching Cases with Status and Bureau/Cell
Create Independent Case if No Relevant Case Exists
6. Coordination and Case Workflow
Request Information • Update Progress
Transfer / Escalate • Record Decisions
+ Link to Existing Complaint
User Links Relevant Case
1. Complaint Registration
Validate Details • Generate Case ID
4. Central Case Database
Institution Profile • Individual Case IDs
Case Links • Status • Action History
2. Institution Matching
AICTE Permanent ID • Institution Records
7. Monitoring and Reports
Pending / Overdue Cases
Institution-Wise and Bureau-Wise Insights
3. Related Case Finder
Check Similar Unresolved Complaints
Notifications
Status Updates • Due-Date Alerts
Evidence Storage
Documents and Attachments